

Math 428 Spring 2024 Homework 8 Solution

6.3 #2

Proof. From Example 6.3.3, $h(z) = \frac{2z}{z-i}$ maps $\partial D_1(0)$ to $\operatorname{Re}(z) = 1$. Clearly $g(z) = z + 1$ maps $\partial D_1(-1)$ to $\partial D_1(0)$. Then the inverse $g^{-1} \circ h^{-1}(z) = \frac{2z}{z-i} - 1$ maps $\operatorname{Re}(z) = 1$ to $\partial D_1(-1)$. $\square$

6.3 #3

Proof. We note $h(1) = \infty$, $h(-1) = i$, and $h(i) = 1 + i$. So $\partial D_1(0)$ is mapped to $\operatorname{Im}(z) = 1$. Then note $h(\frac{1}{2}) = -2i$. This implies $D_1(0)$ is mapped to $\operatorname{Im}(z) < 1$. $\square$

6.3 #4

Proof. We note $h(1) = \infty$, $h(1) = i$, and $h(1 + 2i) = -1 + i$. So $\partial D_1(0)$ is mapped to $\operatorname{Im}(z) = 1$. Then note $h(1 + iy) = -1 + \frac{2i}{y}$ for $y > 0$. This implies $D_1(0)$ is mapped to $\operatorname{Im}(z) > 1$. $\square$

6.3 #6

Proof. Two cases occur when the circle C intersecting $\partial D_1(0)$ lies in the upper or the lower half plane. Suppose C lies in the upper half plane centered at $1 + iy_0$ for some $y_0 > 0$. Then $h(1) = \infty$, $h(1 + i2y_0) = -1 + i\frac{1}{y_0}$, and $h(1 - y_0 + iy_0) = -1 + \frac{1}{y_0} + \frac{i}{y_0}$. So C is mapped to $\operatorname{Im}(z) = \frac{1}{y_0}$. As $h(1 + iy_0) = -1 + \frac{2i}{y_0}$, $\operatorname{int}(C)$ is mapped to $\operatorname{Im}(z) > \frac{1}{y_0}$. So $D_1(0) \setminus \operatorname{int}(C)$ is mapped to $\operatorname{Im}(z) < \frac{1}{y_0}$. But we already know the image of $h(z)$ lie in the right half plane. The other case is similar. $\square$

6.3 #10

Proof. Write the automorphism as $f(z) = u \frac{z-w}{1-\bar{w}z}$ for some u, w such that $|w| < 1$ and $|u| = 1$. Then $f(\frac{1}{2}) = 0$ implies $0 = u \frac{\frac{1}{2}-w}{1-\frac{1}{2}\bar{w}}$. As $|w| < 1$, the denominator is nonzero, which implies $w = \frac{1}{2}$. Now $f(z) = u \frac{z-\frac{1}{2}}{1-\frac{1}{2}\bar{z}}$. It follows $f'(0) = \frac{3}{2}u(-1)(1-\frac{1}{2})^{-2}(-\frac{1}{2}) = \frac{3}{4}i$. So $u = i$. $\square$